

A Corrected Likelihood Reduction for the Wonham-Filter Variant of a Partial-Information Investment-Stopping Problem

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Abstract

This note records a rigorous partial result concerning the hidden Markov-chain extension of the Bayesian-filter reduction in the optimal investment-stopping problem of Xing, Ma and Zheng. It is not a complete solution of the Wonham-filter free-boundary problem. The complete existence, uniqueness, and verification theory for the Wonham free boundary is not proved here. What is proved is the following conditional reduction. Under the same reference-measure idea used in the Bayesian-filter case, the failed endpoint likelihood identity has an exact finite-variation correction:

$$\frac{d\mathbb{P}}{d\mathbb{Q}}\Big|_{\mathcal{F}_t^S} = \frac{1 + \Phi_t}{1 + \varphi} \exp \left\{ \int_0^t (\lambda_h \Phi_s - \lambda_l) \, ds \right\}.$$

This identity gives a pathwise representation of the dual state process in terms of the scalar Wonham likelihood-ratio diffusion and an additive functional. Conditional on the usual threshold free-boundary structure and the early-exercise-premium representation, the boundary equation can then be rewritten without the joint transition law of the two-dimensional process (Y, P) . The remaining kernels are expectations over one scalar diffusion with additive functionals, and, subject to additional transform-integrability assumptions, can be represented by one-dimensional Feynman–Kac equations and a Bromwich inversion. Thus the result is meaningful partial progress, not a full solution.

1 Status of the result

The result below should be read as a corrected-likelihood reduction. It addresses the obstruction that appears when the two-state Bayesian prior in the model of Xing–Ma–Zheng is replaced by a hidden two-state Markov chain. In the Bayesian-filter case the key density is endpoint-only:

$$\frac{d\mathbb{P}}{d\mathbb{Q}}\Big|_{\mathcal{F}_t^S} = \frac{1 + \Phi_t}{1 + \varphi}.$$

In the Markov-switching case this formula is false. Lemma 3.1 gives the correct formula and shows exactly what term is missing.

The status is as follows.

- The corrected density formula and the pathwise scalar representation of Y are proved directly from Ito calculus and Girsanov’s theorem.
- The nonlinear boundary equation is derived conditionally: it assumes that the Wonham-filter stopping problem has the same threshold form and early-exercise-premium identity that are proved in the Bayesian-filter problem.

- The Feynman–Kac and Bromwich formulas are stated with explicit admissibility hypotheses. Establishing those hypotheses for the boundary-dependent functions is part of the remaining work.

Consequently, this note proves partial progress rather than a complete solution of the hidden Markov-chain extension.

2 Model and notation

Let $\alpha_t \in \{h, l\}$ be a continuous-time Markov chain with transition rates

$$h \rightarrow l \text{ at rate } \lambda_h, \quad l \rightarrow h \text{ at rate } \lambda_l, \quad \lambda_h, \lambda_l > 0.$$

The risky asset has unobservable drift $\mu(\alpha_t)$, where

$$\mu_h > \mu_l > r, \quad \theta_i := \frac{\mu_i - r}{\sigma}, \quad i \in \{h, l\}, \quad \Theta := \theta_h - \theta_l = \frac{\mu_h - \mu_l}{\sigma} > 0.$$

Let $\mathcal{F}_t^S$ be the observation filtration and set

$$P_t := \mathbb{P}(\alpha_t = h \mid \mathcal{F}_t^S), \quad \vartheta(p) := p\theta_h + (1 - p)\theta_l = \theta_l + \Theta p.$$

Under the physical measure $\mathbb{P}$, the Wonham filter satisfies

$$dP_t = (\lambda_l(1 - P_t) - \lambda_h P_t) dt + \Theta P_t(1 - P_t) d\widehat{W}_t, \quad (1)$$

where $\widehat{W}$ is the innovation Brownian motion. Starting from $p \in (0, 1)$, the filter remains in $(0, 1)$ on finite time intervals. Define the likelihood-ratio coordinate

$$\Phi_t := \frac{P_t}{1 - P_t}, \quad p(x) := \frac{x}{1 + x}, \quad \varphi := \frac{p}{1 - p}.$$

The dual state process is

$$dY_t = (\beta - r)Y_t dt - \vartheta(P_t)Y_t d\widehat{W}_t. \quad (2)$$

For definiteness, use the utility class appearing in the dimension-reduction part of Xing–Ma–Zheng:

$$\tilde{U}_K(y) = \sum_{j=1}^J -\frac{1}{q_j} y^{q_j} - Ky, \quad q_1 < \dots < q_J < 0, \quad K > 0. \quad (3)$$

For this obstacle, the early-exercise-premium integrand is

$$\phi(y, p) := (\beta - \mathcal{L}_{Y,P})\tilde{U}_K(y) = \sum_{j=1}^J A_j(p)y^{q_j} - rKy, \quad (4)$$

where $\mathcal{L}_{Y,P}$ is the $\mathbb{P}$ -generator of (Y, P) and

$$A_j(p) := \frac{1}{2}\vartheta(p)^2(q_j - 1) + \left(1 - \frac{1}{q_j}\right)\beta - r. \quad (5)$$

Assumption 2.1 (Free-boundary and premium representation). The Wonham-filter dual stopping problem has a continuous positive threshold boundary b such that the stopping set is

$$\{(t, y, p) : 0 < y \leq b(t, p)\}. \quad (6)$$

At each boundary point, the early-exercise-premium identity holds:

$$0 = \mathbb{E}_{t, b(t, p), p}^{\mathbb{P}} \left[\int_t^T e^{-\beta(s-t)} \phi(Y_s, P_s) \mathbf{1}_{\{Y_s > b(s, P_s)\}} ds \right]. \quad (7)$$

Moreover,

$$\lim_{t \uparrow T} b(t, p) = \ell(p), \quad (8)$$

where $\ell(p) > 0$ is the unique positive root of $\phi(y, p) = 0$. All integrability needed for (7), the change of measure, and Fubini's theorem in the derivation below is assumed.

This assumption is the main conditional point of the note. The results below do not prove Assumption 2.1; they prove what follows from it once the likelihood correction is made.

3 The corrected likelihood identity

Introduce the reference measure $\mathbb{Q}$ by

$$dW_t = d\widehat{W}_t + \Theta P_t dt. \quad (9)$$

Equivalently,

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t^S} = \mathcal{E} \left(- \int_0^t \Theta P_s d\widehat{W}_s \right).$$

Novikov's condition is immediate because $0 \leq P_t \leq 1$. Under $\mathbb{Q}$, W is an $\mathcal{F}^S$ -Brownian motion. Applying Ito's formula to $\Phi = P/(1 - P)$, using (1) and (9), gives

$$d\Phi_t = B(\Phi_t) dt + \Theta \Phi_t dW_t, \quad (10)$$

where

$$B(x) := (\lambda_l - \lambda_h x)(1 + x), \quad x > 0. \quad (11)$$

Set

$$r_0(x) := \lambda_h x - \lambda_l, \quad R_t := \int_0^t r_0(\Phi_s) ds. \quad (12)$$

Lemma 3.1 (Corrected density). *For every $t \in [0, T]$, conditional on $\Phi_0 = \varphi$,*

$$F_t := \frac{d\mathbb{P}}{d\mathbb{Q}} \Big|_{\mathcal{F}_t^S} = \frac{1 + \Phi_t}{1 + \varphi} e^{R_t}. \quad (13)$$

Equivalently,

$$dF_t = \Theta p(\Phi_t) F_t dW_t, \quad F_0 = 1. \quad (14)$$

Proof. By (10),

$$\begin{aligned} d(e^{R_t}(1 + \Phi_t)) &= e^{R_t} d\Phi_t + e^{R_t}(1 + \Phi_t)r_0(\Phi_t) dt \\ &= e^{R_t} [B(\Phi_t) + (1 + \Phi_t)r_0(\Phi_t)] dt + e^{R_t} \Theta \Phi_t dW_t. \end{aligned}$$

The drift vanishes because

$$B(x) + (1+x)r_0(x) = (\lambda_l - \lambda_h x)(1+x) + (1+x)(\lambda_h x - \lambda_l) = 0.$$

After division by $1 + \varphi$, this gives

$$d\left(\frac{e^{R_t}(1 + \Phi_t)}{1 + \varphi}\right) = \Theta \frac{\Phi_t}{1 + \Phi_t} \frac{e^{R_t}(1 + \Phi_t)}{1 + \varphi} dW_t.$$

This is (14). Since $p(\Phi_t) = P_t \in [0, 1]$, the stochastic exponential solving (14) is a true martingale. It is the inverse density for the Girsanov change (9); hence (13) follows. $\square$

Remark 3.2. If $\lambda_h = \lambda_l = 0$, then $R_t \equiv 0$, and Lemma 3.1 reduces to the endpoint-only Bayesian-filter identity $F_t = (1 + \Phi_t)/(1 + \varphi)$. For positive switching intensities, the finite-variation factor e^{R_t} is the missing term.

4 A pathwise scalar representation of the dual process

Under $\mathbb{Q}$, define the auxiliary process

$$d\mathcal{Y}_t = (\beta - r)\mathcal{Y}_t dt - \theta_l \mathcal{Y}_t dW_t, \quad \mathcal{Y}_0 = y. \quad (15)$$

Then

$$Y_t = \frac{\mathcal{Y}_t}{F_t}. \quad (16)$$

Indeed, using (14), Ito's formula shows that the quotient has, under $\mathbb{Q}$, diffusion coefficient $-\vartheta(P_t)Y_t$ and drift $(\beta - r + \Theta P_t \vartheta(P_t))Y_t$, which is exactly (2) rewritten with $d\widehat{W}_t = dW_t - \Theta P_t dt$.

Set

$$a := \frac{\theta_l}{\Theta}, \quad h(x) := x^a(1+x), \quad \zeta := \beta - r - \frac{1}{2}\theta_l^2 - \frac{1}{2}\theta_l\Theta, \quad (17)$$

and

$$c_0(x) := a \frac{B(x)}{x} - r_0(x) = \frac{\theta_l}{\Theta} \frac{(\lambda_l - \lambda_h x)(1+x)}{x} - (\lambda_h x - \lambda_l), \quad (18)$$

with

$$C_t := \int_0^t c_0(\Phi_s) ds. \quad (19)$$

Lemma 4.1 (Pathwise representation). *For every $t \in [0, T]$, conditional on $Y_0 = y$ and $\Phi_0 = \varphi$,*

$$Y_t = y \frac{h(\varphi)}{h(\Phi_t)} \exp\{\zeta t + C_t\}. \quad (20)$$

Proof. The explicit solution of (15) is

$$\mathcal{Y}_t = y \exp\left\{\left(\beta - r - \frac{1}{2}\theta_l^2\right)t - \theta_l W_t\right\}.$$

From (10),

$$d \log \Phi_t = \left(\frac{B(\Phi_t)}{\Phi_t} - \frac{1}{2}\Theta^2\right) dt + \Theta dW_t.$$

Hence

$$W_t = \frac{1}{\Theta} \left[\log \frac{\Phi_t}{\varphi} - \int_0^t \frac{B(\Phi_s)}{\Phi_s} ds + \frac{1}{2}\Theta^2 t \right].$$

Substituting this expression into $\mathcal{Y}_t$, dividing by (13), and collecting terms gives (20). The endpoint terms combine to $h(\varphi)/h(\Phi_t)$, the deterministic exponent is ζt , and the remaining integral is C_t . $\square$

5 Scalar additive-functional kernels

Let

$$\mathcal{L}_\Phi f(x) := B(x)f'(x) + \frac{1}{2}\Theta^2 x^2 f''(x), \quad x > 0, \quad (21)$$

be the $\mathbb{Q}$ -generator of Φ .

Definition 5.1 (Feynman–Kac kernel). For a measurable $G : (0, \infty) \rightarrow \mathbb{C}$ and complex ω, ν for which the expectation is finite, define

$$(\mathcal{K}_u^{\omega, \nu} G)(\varphi) := \mathbb{E}_\varphi^\mathbb{Q}[\exp\{\omega R_u + \nu C_u\} G(\Phi_u)]. \quad (22)$$

Under standard Feynman–Kac hypotheses, $v(u, x) = (\mathcal{K}_u^{\omega, \nu} G)(x)$ solves

$$\begin{cases} \partial_u v = \mathcal{L}_\Phi v + (\omega r_0 + \nu c_0)v, & u > 0, \ x > 0, \\ v(0, x) = G(x), & x > 0. \end{cases} \quad (23)$$

Definition 5.2 (Truncated kernel). For a Borel threshold $\kappa : (0, \infty) \rightarrow \mathbb{R}$, define

$$(\mathcal{T}_u^{\omega, \nu, \kappa} G)(\varphi) := \mathbb{E}_\varphi^\mathbb{Q}\left[e^{\omega R_u + \nu C_u} G(\Phi_u) \mathbf{1}_{\{C_u > \kappa(\Phi_u)\}}\right], \quad (24)$$

whenever the expectation is finite.

Proposition 5.3 (Bromwich representation, conditional on admissibility). *Fix $u > 0$, $\varphi > 0$, G , κ , ω , and ν . Suppose there exists $\alpha > 0$ such that the relevant transforms exist on the line $\nu + \alpha + i\mathbb{R}$, the Fourier inversion of the Heaviside function is valid at $C_u - \kappa(\Phi_u)$ almost surely, and the expectation/interchange limits are justified. Then*

$$(\mathcal{T}_u^{\omega, \nu, \kappa} G)(\varphi) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\left(\mathcal{K}_u^{\omega, \nu + \alpha + i\xi} \left[G(\cdot) e^{-(\alpha + i\xi)\kappa(\cdot)}\right]\right)(\varphi)}{\alpha + i\xi} d\xi. \quad (25)$$

Proof. For real z, k away from the discontinuity $z = k$, the inverse Laplace representation of the Heaviside function gives

$$e^{\nu z} \mathbf{1}_{\{z > k\}} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{(\nu + \alpha + i\xi)z} e^{-(\alpha + i\xi)k}}{\alpha + i\xi} d\xi, \quad \alpha > 0.$$

Apply this with $z = C_u$ and $k = \kappa(\Phi_u)$, multiply by $e^{\omega R_u} G(\Phi_u)$, take expectations, and use the assumed interchange of the Fourier integral and expectation. The resulting expression is exactly (25). $\square$

Remark 5.4. Proposition 5.3 is not a closed analytic theorem for the present stopping problem, because $c_0(x)$ is singular near 0 and unbounded as $x \rightarrow \infty$, and because κ depends on the unknown boundary. These analytic issues are part of the remaining work. The exact boundary equation in the next section uses the probabilistic definition (24); the transform formula is an additional computational representation when admissibility is available.

6 The conditional Volterra equation

Fix $t < T$ and $\varphi > 0$. Write

$$p_\varphi := p(\varphi), \quad y_b := b(t, p_\varphi). \quad (26)$$

For $0 \leq u \leq T - t$ and $x > 0$, define

$$H_{t,u}^{b,\varphi}(x) := y_b \frac{h(\varphi)}{h(x)} e^{\zeta u}. \quad (27)$$

If $(Y_0, \Phi_0) = (y_b, \varphi)$, Lemma 4.1 gives

$$Y_u = H_{t,u}^{b,\varphi}(\Phi_u) e^{C_u}.$$

Therefore the continuation event is

$$\{Y_u > b(t+u, p(\Phi_u))\} = \{C_u > \kappa_{t,u}^{b,\varphi}(\Phi_u)\}, \quad (28)$$

where

$$\kappa_{t,u}^{b,\varphi}(x) := \log \left(\frac{b(t+u, p(x))}{H_{t,u}^{b,\varphi}(x)} \right) = \log \left(\frac{b(t+u, p(x))h(x)}{b(t, p_\varphi)h(\varphi)e^{\zeta u}} \right). \quad (29)$$

Define

$$G_{j,t,u}^{b,\varphi}(x) := \frac{1+x}{1+\varphi} A_j(p(x)) \left(H_{t,u}^{b,\varphi}(x) \right)^{q_j}, \quad j = 1, \dots, J, \quad (30)$$

$$G_{K,t,u}^{b,\varphi}(x) := \frac{1+x}{1+\varphi} r K H_{t,u}^{b,\varphi}(x). \quad (31)$$

Theorem 6.1 (Necessary scalar-kernel equation). *Assume Assumption 2.1. Then the true Wonham-filter boundary satisfies, for every $t < T$ and $\varphi > 0$,*

$$\boxed{0 = \int_0^{T-t} e^{-\beta u} \left[\sum_{j=1}^J \left(\mathcal{T}_u^{1,q_j, \kappa_{t,u}^{b,\varphi}} G_{j,t,u}^{b,\varphi} \right) (\varphi) - \left(\mathcal{T}_u^{1,1, \kappa_{t,u}^{b,\varphi}} G_{K,t,u}^{b,\varphi} \right) (\varphi) \right] du.} \quad (32)$$

Thus, conditional on the free-boundary hypothesis, the original boundary equation can be written using only expectations over the scalar diffusion Φ and the additive functionals R and C .

Proof. By time homogeneity, restart the process at time t and write $u = s - t$. At the boundary point $(t, b(t, p_\varphi), p_\varphi)$, (7) becomes

$$0 = \mathbb{E}_{y_b, \varphi}^{\mathbb{P}} \left[\int_0^{T-t} e^{-\beta u} \phi(Y_u, p(\Phi_u)) \mathbf{1}_{\{Y_u > b(t+u, p(\Phi_u))\}} du \right].$$

Change measure from $\mathbb{P}$ to $\mathbb{Q}$ and use Lemma 3.1:

$$0 = \int_0^{T-t} e^{-\beta u} \mathbb{E}_{y_b, \varphi}^{\mathbb{Q}} \left[F_u \phi(Y_u, p(\Phi_u)) \mathbf{1}_{\{Y_u > b(t+u, p(\Phi_u))\}} \right] du.$$

By Lemmas 3.1 and 4.1,

$$F_u = \frac{1 + \Phi_u}{1 + \varphi} e^{R_u}, \quad Y_u = H_{t,u}^{b,\varphi}(\Phi_u) e^{C_u}.$$

Using (4), (30), and (31),

$$F_u \phi(Y_u, p(\Phi_u)) = \sum_{j=1}^J e^{R_u + q_j C_u} G_{j,t,u}^{b,\varphi}(\Phi_u) - e^{R_u + C_u} G_{K,t,u}^{b,\varphi}(\Phi_u).$$

The event in the indicator is exactly (28). Substituting these identities and applying Definition 5.2 gives (32). $\square$

Remark 6.2 (No converse is claimed). Theorem 6.1 proves a necessary equation for the true boundary under Assumption 2.1. It does not prove that every positive continuous solution of (32) is the true boundary. A converse would require a separate verification theorem: construction of a candidate value function, enough regularity to justify the variational inequality, boundary and terminal conditions, and a comparison or uniqueness argument. These steps are not supplied here.

7 Endpoint-only reductions are impossible with positive switching

The finite-variation correction in Lemma 3.1 is forced by the switching drift. The next proposition rules out an endpoint-only density of the Bayesian type in a natural family of reference measures.

Proposition 7.1 (No endpoint-only density of the Bayesian type). *Fix a constant θ_0 and define a reference measure by*

$$dW_t = d\widehat{W}_t + (\vartheta(P_t) - \theta_0) dt.$$

Assume $\lambda_h, \lambda_l > 0$. There do not exist a constant $c \in \mathbb{R}$ and a positive C^2 function $f : (0, \infty) \rightarrow (0, \infty)$ such that, for every initial $\varphi > 0$,

$$\frac{d\mathbb{P}}{d\mathbb{Q}} \Big|_{\mathcal{F}_t^S} = e^{ct} \frac{f(\Phi_t)}{f(\varphi)} \quad (33)$$

for all t .

Proof. Put

$$\delta := \frac{\theta_l - \theta_0}{\Theta}.$$

The martingale coefficient of the density $d\mathbb{P}/d\mathbb{Q}$ under this reference measure is

$$\vartheta(p(\Phi_t)) - \theta_0 = \theta_l - \theta_0 + \frac{\Theta \Phi_t}{1 + \Phi_t} = \Theta \left(\delta + \frac{\Phi_t}{1 + \Phi_t} \right).$$

If (33) held, matching the martingale parts in Ito's formula would force

$$\Theta x \frac{f'(x)}{f(x)} = \Theta \left(\delta + \frac{x}{1+x} \right),$$

so

$$f(x) = Cx^\delta(1+x), \quad C > 0. \quad (34)$$

Under the same reference measure, the diffusion coefficient of Φ is Θx and its drift is

$$B(x) - \Theta^2 \delta x.$$

From (34),

$$\frac{f''(x)}{f(x)} = \frac{\delta(\delta-1)}{x^2} + \frac{2\delta}{x(1+x)}.$$

Consequently the logarithmic generator ratio is

$$\begin{aligned}\frac{\mathcal{L}_\Phi f(x)}{f(x)} &= (B(x) - \Theta^2 \delta x) \left(\frac{\delta}{x} + \frac{1}{1+x} \right) + \frac{1}{2} \Theta^2 x^2 \left(\frac{\delta(\delta-1)}{x^2} + \frac{2\delta}{x(1+x)} \right) \\ &= B(x) \left(\frac{\delta}{x} + \frac{1}{1+x} \right) - \frac{1}{2} \Theta^2 \delta(\delta+1).\end{aligned}$$

The second term is constant. The first term equals

$$(\lambda_l - \lambda_h x)(1+x) \left(\frac{\delta}{x} + \frac{1}{1+x} \right) = (\lambda_l - \lambda_h x) \left(\frac{\delta}{x} + \delta + 1 \right).$$

If $\delta \neq 0$, this contains the nonconstant term $\delta \lambda_l / x$. If $\delta = 0$, it becomes $\lambda_l - \lambda_h x$, which is nonconstant because $\lambda_h > 0$. Therefore $(\mathcal{L}_\Phi f)/f$ is never constant. Hence no choice of c can make $e^{ct} f(\Phi_t)/f(\varphi)$ a density martingale. This contradiction proves the claim. $\square$

Remark 7.2. For the reference measure used in Lemma 3.1, $\theta_0 = \theta_l$, so $\delta = 0$. The best endpoint factor is then $1 + \Phi_t$, and the remaining nonconstant drift $\lambda_l - \lambda_h \Phi_t$ is exactly cancelled by the additive functional $R_t = \int_0^t (\lambda_h \Phi_s - \lambda_l) ds$.

8 Bayesian-filter limit

If $\lambda_h = \lambda_l = 0$, then

$$B(x) = 0, \quad r_0(x) = 0, \quad c_0(x) = 0, \quad R_t = C_t = 0.$$

The likelihood ratio satisfies $d\Phi_t = \Theta \Phi_t dW_t$, and the density in Lemma 3.1 becomes the endpoint-only expression of the Bayesian-filter model. In (32), the indicator threshold no longer involves an independent additive functional: the continuation event is determined by the endpoint Φ_u alone. Thus the additional transform machinery collapses, formally recovering the dimension-reduced Bayesian equation. This is a consistency check for the corrected density and for the conditional reduction.

9 What remains for a full solution

The note proves the likelihood correction and a conditional one-dimensional reduction, but a full solution of the Wonham-filter extension would still require at least the following steps.

1. **Free-boundary structure.** Prove, for the Wonham generator, that the dual stopping region is a threshold set $\{0 < y \leq b(t, p)\}$, with sufficient regularity and the terminal limit (8).
2. **Verification and uniqueness.** Prove that a solution of the nonlinear equation (32), together with the terminal condition and appropriate monotonicity properties, actually yields the value function and the optimal stopping time.
3. **Transform analysis.** Establish exponential moment bounds and strip analyticity for the additive functionals R and C , justify (25) for the boundary-dependent functions that occur in (32), and handle the singular/unbounded potential c_0 in (23).
4. **Numerical convergence.** Give a stable backward-recursive discretization of (32) and prove convergence of the PDE, Fourier, time, and boundary iterations.

These are substantial mathematical tasks. The present result should therefore be cited as a corrected-likelihood reduction and a necessary scalar boundary equation, not as a completed solution of the hidden Markov-chain optimal stopping problem.

References

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